

Effect of a WNBA Team on Girls' Sports Participation

Tate Mason

Invalid Date

Introduction

This file serves as the data analysis portion of my project. I will be using NFHS data and WNBA data sourced from Kaggle to analyze the effect of a WNBA team on the youth sports participation of girls in a given state. I will be focusing on Basketball, Track, and Softball, as these are the most available sports in WFHS.

Data Cleaning

NFHS data was cleaned by Claude Code, scraping PDF's and converting to CSV's. WNBA data is downloaded clean. Still, I want to make sure things are formatted correctly.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(readr)
library(ggplot2)
library(knitr)
library(kableExtra)
```

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

group_rows

```
library(psych)
```

Attaching package: 'psych'

The following objects are masked from 'package:ggplot2':

%+%, alpha

```
df_wnba <- read_csv("~/SchoolWork/Y2S2/Health/ResProj/Data/WNBA_Standings.csv")
```

Rows: 352 Columns: 13

-- Column specification -----

Delimiter: ","

chr (8): TEAM, CONF, DIV, HOME, ROAD, OT, LAST 10, STREAK

dbl (5): RANK, W, L, WIN%, YEAR

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
df_nfhs_wide <- read_csv("~/SchoolWork/Y2S2/Health/ResProj/Data/NFHS/nfhs_processed/state_sp")
```

Rows: 1715 Columns: 26

-- Column specification -----

Delimiter: ","

chr (3): survey_year, gender, state

dbl (23): Baseball, Basketball, Bowling, Competitive Spirit Squads, Cross Co...

- i Use ``spec()`` to retrieve the full column specification for this data.
- i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
# Looking at WNBA data
```

```
dummy <- rep(c(1,0), length.out = nrow(df_wnba))
df_wnba <- df_wnba %>%
  rename_with(tolower) %>%
  mutate(east = dummy,
         west = 1 - dummy) %>%
  filter(year <= 2019)
head(df_wnba)
```

```
# A tibble: 6 x 15
```

	rank	team	w	l	`win%`	conf	div	home	road	ot	`last 10`	streak
	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	1	NEW Y~	17	11	0.607	5-3	0-0	10-4	7-7	0-0	3-7	W 1
2	1	HOUST~	18	10	0.643	12-4	0-0	9-5	9-5	0-0	7-3	L 2
3	2	CHARL~	15	13	0.536	4-4	0-0	12-2	3-11	0-0	5-5	W 1
4	2	PHOEN~	16	12	0.571	10-6	0-0	11-3	5-9	0-0	7-3	W 3
5	3	CLEVE~	15	13	0.536	3-5	0-0	8-6	7-7	0-0	5-5	L 1
6	3	LOS A~	14	14	0.5	9-7	0-0	8-6	6-8	0-0	6-4	L 1

i 3 more variables: year <dbl>, east <dbl>, west <dbl>

```
summary(df_wnba)
```

rank	team	w	l
Min. : 1.000	Length:292	Min. : 3.00	Min. : 3.00
1st Qu.: 2.000	Class :character	1st Qu.:13.00	1st Qu.:13.00
Median : 4.000	Mode :character	Median :17.00	Median :16.00
Mean : 4.248		Mean :16.64	Mean :16.64
3rd Qu.: 6.000		3rd Qu.:20.00	3rd Qu.:20.00
Max. :12.000		Max. :29.00	Max. :31.00
NA's :14			
win%	conf	div	home
Min. :0.0880	Length:292	Length:292	Length:292
1st Qu.:0.4045	Class :character	Class :character	Class :character
Median :0.5000	Mode :character	Mode :character	Mode :character
Mean :0.5000			
3rd Qu.:0.6180			

Max. :0.9000

road	ot	last 10	streak
Length:292	Length:292	Length:292	Length:292
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

year	east	west
Min. :1997	Min. :0.0	Min. :0.0
1st Qu.:2002	1st Qu.:0.0	1st Qu.:0.0
Median :2008	Median :0.5	Median :0.5
Mean :2008	Mean :0.5	Mean :0.5
3rd Qu.:2013	3rd Qu.:1.0	3rd Qu.:1.0
Max. :2019	Max. :1.0	Max. :1.0

```
# Looking at NFHS data

df_nfhs_wide <- df_nfhs_wide %>%
  mutate(
    gender = case_when(
      gender == "Boys" ~ 0,
      gender == "Girls" ~ 1,
      TRUE ~ NA_real_
    ),
    state = case_when(
      state == "Alabama" ~ 1,
      state == "Alaska" ~ 2,
      state == "Arizona" ~ 3,
      state == "Arkansas" ~ 4,
      state == "California" ~ 5,
      state == "Colorado" ~ 6,
      state == "Connecticut" ~ 7,
      state == "Delaware" ~ 8,
      state == "Florida" ~ 9,
      state == "Georgia" ~ 10,
      state == "Hawaii" ~ 11,
      state == "Idaho" ~ 12,
      state == "Illinois" ~ 13,
```

```

state == "Indiana" ~ 14,
state == "Iowa" ~ 15,
state == "Kansas" ~ 16,
state == "Kentucky" ~ 17,
state == "Louisiana" ~ 18,
state == "Maine" ~ 19,
state == "Maryland" ~ 20,
state == "Massachusetts" ~ 21,
state == "Michigan" ~ 22,
state == "Minnesota" ~ 23,
state == "Mississippi" ~ 24,
state == "Missouri" ~ 25,
state == "Montana" ~ 26,
state == "Nebraska" ~ 27,
state == "Nevada" ~ 28,
state == "New Hampshire" ~ 29,
state == "New Jersey" ~ 30,
state == "New Mexico" ~ 31,
state == "New York" ~ 32,
state == "North Carolina" ~ 33,
state == "North Dakota" ~ 34,
state == "Ohio" ~ 35,
state == "Oklahoma" ~ 36,
state == "Oregon" ~ 37,
state == "Pennsylvania" ~ 38,
state == "Rhode Island" ~ 39,
state == "South Carolina" ~ 40,
state == "South Dakota" ~ 41,
state == "Tennessee" ~ 42,
state == "Texas" ~ 43,
state == "Utah" ~ 44,
state == "Vermont" ~ 45,
state == "Virginia" ~ 46,
state == "Washington" ~ 47,
state == "West Virginia" ~ 48,
state == "Wisconsin" ~ 49,
state == "Wyoming" ~ 50,
TRUE ~ NA_real_
),
year = case_when(
  survey_year == "1993-94" ~ 1993,
  survey_year == "1994-95" ~ 1994,

```

```

    survey_year == "1995-96" ~ 1995,
    survey_year == "1996-97" ~ 1996,
    survey_year == "1997-98" ~ 1997,
    survey_year == "1998-99" ~ 1998,
    survey_year == "1999-00" ~ 1999,
    survey_year == "2000-01" ~ 2000,
    survey_year == "2001-02" ~ 2001,
    survey_year == "2002-03" ~ 2002,
    survey_year == "2003-04" ~ 2003,
    survey_year == "2004-05" ~ 2004,
    survey_year == "2005-06" ~ 2005,
    survey_year == "2006-07" ~ 2006,
    survey_year == "2007-08" ~ 2007,
    survey_year == "2008-09" ~ 2008,
    survey_year == "2009-10" ~ 2009,
    survey_year == "2010-11" ~ 2010,
    survey_year == "2011-12" ~ 2011,
    survey_year == "2013-14" ~ 2013,
    survey_year == "2015-16" ~ 2015,
    survey_year == "2017-18" ~ 2017,
    survey_year == "2018-19" ~ 2019,
    TRUE ~ NA_real_
  )
)

head(df_nfhs_wide)

```

```

# A tibble: 6 x 27
  survey_year gender state Baseball Basketball Bowling Competitive Spirit Squa~1
  <chr>         <dbl> <dbl>    <dbl>      <dbl>    <dbl>      <dbl>      <dbl>
1 2009-10         0     1    8966      11899     NA              NA
2 2010-11         0     1    8827      11755     NA              NA
3 2011-12         0     1    8717      11630     NA              NA
4 2012-13         0     1    8913      11720     NA              NA
5 2013-14         0     1    8822      11731     NA              NA
6 2014-15         0     1   12080      12860    171              NA
# i abbreviated name: 1: `Competitive Spirit Squads`
# i 20 more variables: `Cross Country` <dbl>, `Field Hockey` <dbl>,
#   `Football 11-player` <dbl>, Golf <dbl>, Gymnastics <dbl>,
#   `Ice Hockey` <dbl>, Lacrosse <dbl>, Riflery <dbl>, `Skiing Alpine` <dbl>,
#   `Skiing Cross Country` <dbl>, Soccer <dbl>, `Softball Fast Pitch` <dbl>,
#   `Softball Slow Pitch` <dbl>, `Swimming and Diving` <dbl>, Tennis <dbl>,

```

```
# `Track and Field Indoor` <dbl>, `Track and Field Outdoor` <dbl>, ...
```

```
summary(df_nfhs_wide)
```

survey_year	gender	state	Baseball
Length:1715	Min. :0.0000	Min. : 1.00	Min. : 71
Class :character	1st Qu.:0.0000	1st Qu.:13.00	1st Qu.: 2706
Mode :character	Median :1.0000	Median :26.00	Median : 7689
	Mean :0.7079	Mean :25.53	Mean : 9678
	3rd Qu.:1.0000	3rd Qu.:38.00	3rd Qu.:12895
	Max. :1.0000	Max. :50.00	Max. :48706
		NA's :21	NA's :1239
Basketball	Bowling	Competitive Spirit Squads	Cross Country
Min. : 11	Min. : 2.00	Min. : 1	Min. : 65
1st Qu.: 3066	1st Qu.: 85.75	1st Qu.: 179	1st Qu.: 1057
Median : 6626	Median : 388.50	Median : 508	Median : 2752
Mean : 8803	Mean : 903.83	Mean : 1862	Mean : 4196
3rd Qu.:10494	3rd Qu.:1312.00	3rd Qu.: 2277	3rd Qu.: 5364
Max. :74675	Max. :5545.00	Max. :28381	Max. :32951
NA's :227	NA's :1003	NA's :706	NA's :367
Field Hockey	Football 11-player	Golf	Gymnastics
Min. : 2	Min. : 1	Min. : 1	Min. : 1.0
1st Qu.: 567	1st Qu.: 46	1st Qu.: 497	1st Qu.: 105.0
Median : 1142	Median : 391	Median : 1199	Median : 270.0
Mean : 2502	Mean : 10606	Mean : 2069	Mean : 673.8
3rd Qu.: 3731	3rd Qu.: 15367	3rd Qu.: 2678	3rd Qu.: 974.8
Max. :18159	Max. :168680	Max. :30222	Max. :9879.0
NA's :1278	NA's :727	NA's :318	NA's :973
Ice Hockey	Lacrosse	Riflery	Skiing Alpine
Min. : 2.0	Min. : 6	Min. : 3.0	Min. : 1.0
1st Qu.: 83.0	1st Qu.: 1036	1st Qu.: 26.0	1st Qu.: 79.0
Median : 223.5	Median : 2502	Median : 99.0	Median : 183.0
Mean : 913.4	Mean : 3473	Mean :191.9	Mean : 301.6
3rd Qu.: 849.5	3rd Qu.: 4952	3rd Qu.:263.0	3rd Qu.: 375.5
Max. :7520.0	Max. :16206	Max. :944.0	Max. :3119.0
NA's :1193	NA's :1136	NA's :1600	NA's :996
Skiing Cross Country	Soccer	Softball Fast Pitch	Softball Slow Pitch
Min. : 2.0	Min. : 10	Min. : 1	Min. : 10.0
1st Qu.: 113.0	1st Qu.: 1750	1st Qu.: 1086	1st Qu.: 159.8
Median : 282.0	Median : 4259	Median : 4425	Median : 361.0
Mean : 637.7	Mean : 6849	Mean : 5945	Mean : 1871.1
3rd Qu.: 689.5	3rd Qu.: 8828	3rd Qu.: 8086	3rd Qu.: 1206.0

Max. :25157.0	Max. :54996	Max. :40214	Max. :35604.0
NA's :1216	NA's :357	NA's :886	NA's :1101
Swimming and Diving	Tennis	Track and Field Indoor	
Min. : 8.0	Min. : 49.0	Min. : 15.0	
1st Qu.: 625.5	1st Qu.: 872.5	1st Qu.: 252.0	
Median : 1773.0	Median : 2275.0	Median : 450.5	
Mean : 2990.4	Mean : 4922.6	Mean : 1687.9	
3rd Qu.: 4068.0	3rd Qu.: 4692.0	3rd Qu.: 1469.8	
Max. :31161.0	Max. :865437.0	Max. :16001.0	
NA's :524	NA's :336	NA's :549	
Track and Field Outdoor	Volleyball	Wrestling	year
Min. : 1	Min. : 1.0	Min. : 22	Min. :1993
1st Qu.: 2383	1st Qu.: 355.2	1st Qu.: 1531	1st Qu.:2001
Median : 5677	Median : 2658.0	Median : 4359	Median :2008
Mean : 8472	Mean : 5182.4	Mean : 5221	Mean :2007
3rd Qu.:10275	3rd Qu.: 7114.0	3rd Qu.: 7806	3rd Qu.:2013
Max. :74522	Max. :50501.0	Max. :27833	Max. :2019
NA's :410	NA's :429	NA's :1231	NA's :302

Data Analysis

```
unique(df_wnba$year)
```

```
[1] 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011
[16] 2012 2013 2014 2015 2016 2017 2018 2019
```

```
# Map teams to states
team_state <- tribble(
  ~team,                ~state,
  "NEW YORK LIBERTY",    "new york",
  "HOUSTON COMETS",      "texas",
  "CHARLOTTE STING",     "north carolina",
  "PHOENIX MERCURY",     "arizona",
  "CLEVELAND ROCKERS",   "ohio",
  "LOS ANGELES SPARKS",  "california",
  "SACRAMENTO MONARCHS", "california",
  "UTAH STARZZ",         "utah",
  "WASHINGTON MYSTICS",  "virginia",
  "DETROIT SHOCK",       "michigan",
```



```

"ORLANDO MIRACLE",      "florida",
"MINNESOTA LYNX",       "minnesota",
"MIAMI SOL",            "florida",
"PORTLAND FIRE",        "oregon",
"INDIANA FEVER",        "indiana",
"SEATTLE STORM",        "washington",
"CONNECTICUT SUN",       "connecticut",
"SAN ANTONIO SILVER STARS", "texas",
"SAN ANTONIO STARS",     "texas",
"ATLANTA DREAM",        "georgia",
"CHICAGO SKY",          "illinois",
"TULSA SHOCK",          "oklahoma",
"DALLAS WINGS",         "texas",
"LAS VEGAS ACES",       "nevada"
)

# Join state onto standings, get years each state had a team
state_years <- df_wnba %>%
  left_join(team_state, by = "team") %>%
  filter(!is.na(state)) %>%
  distinct(state, year)

# US map data
us_states <- map_data("state")

# Join map data with state_years
map_df <- us_states %>%
  left_join(state_years, by = c("region" = "state"))

```

Warning in left_join(., state_years, by = c(region = "state")): Detected an unexpected many-to-many relationship between `x` and `y`.

- i Row 203 of `x` matches multiple rows in `y`.
- i Row 4 of `y` matches multiple rows in `x`.
- i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

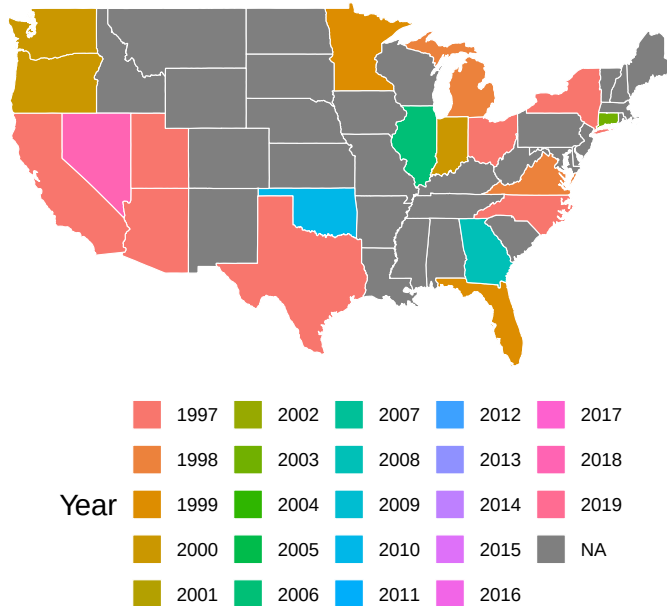
```

ggplot() +
  geom_polygon(data = map_df,
    aes(x = long, y = lat, group = group, fill = factor(year)),
    color = "white", linewidth = 0.2) +
  coord_fixed(1.3) +

```

```
labs(title = "States with WNBA Teams by Year", fill = "Year") +
theme_void() +
theme(legend.position = "bottom",
      legend.key.size = unit(0.4, "cm"),
      legend.text = element_text(size = 7))
```

States with WNBA Teams by Year



Now time series of participation rates in basketball, track, and softball for girls

```
wnba_states <- df_wnba %>%
  select(team, year) %>%
  left_join(team_state, by = "team") %>%
  filter(!is.na(state)) %>%
  mutate(state_num = case_when(
    state == "arizona" ~ 3,
    state == "california" ~ 5,
    state == "connecticut" ~ 7,
    state == "florida" ~ 9,
    state == "georgia" ~ 10,
    state == "illinois" ~ 13,
    state == "indiana" ~ 14,
    state == "michigan" ~ 22,
    state == "minnesota" ~ 23,
    state == "nevada" ~ 28,
```

```

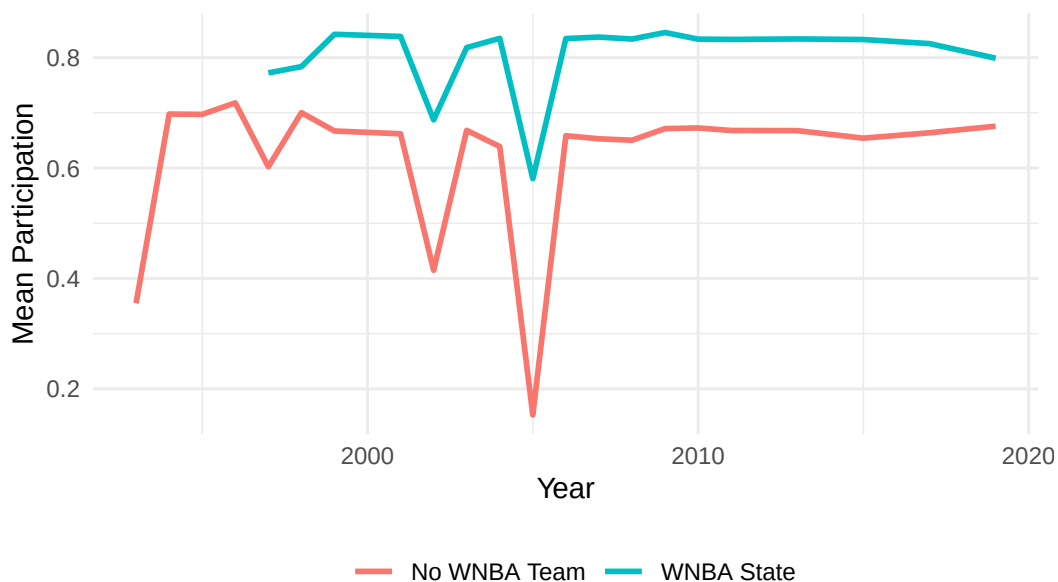
state == "new york" ~ 32,
state == "north carolina" ~ 33,
state == "ohio" ~ 35,
state == "oklahoma" ~ 36,
state == "oregon" ~ 37,
state == "texas" ~ 43,
state == "utah" ~ 44,
state == "virginia" ~ 46,
state == "washington" ~ 47
)) %>%
distinct(state_num, year) %>%
filter(!is.na(state_num))

participation_df <- df_nfhs_wide %>%
  filter(gender == 1) %>%
  select(state, year, Basketball) %>%
  mutate(total = rowSums(across(where(is.numeric)), na.rm = TRUE)) %>%
  mutate(basketball_share = Basketball / total) %>%
  left_join(wnba_states %>% mutate(has_wnba = 1),
            by = c("state" = "state_num", "year")) %>%
  mutate(has_wnba = replace_na(has_wnba, 0),
         group = ifelse(has_wnba == 1, "WNBA State", "No WNBA Team")) %>%
  group_by(group, year) %>%
  summarise(participation_rate = mean(basketball_share, na.rm = TRUE), .groups = "drop") %>%
  filter(!is.na(year))

ggplot(participation_df, aes(x = year, y = participation_rate, color = group, group = group)) +
  geom_line(linewidth = 1) +
  labs(title = "Girls Basketball Participation (Share of Total Participation): WNBA vs Non-WNBA",
       x = "Year", y = "Mean Participation", color = "") +
  theme_minimal() +
  theme(legend.position = "bottom")

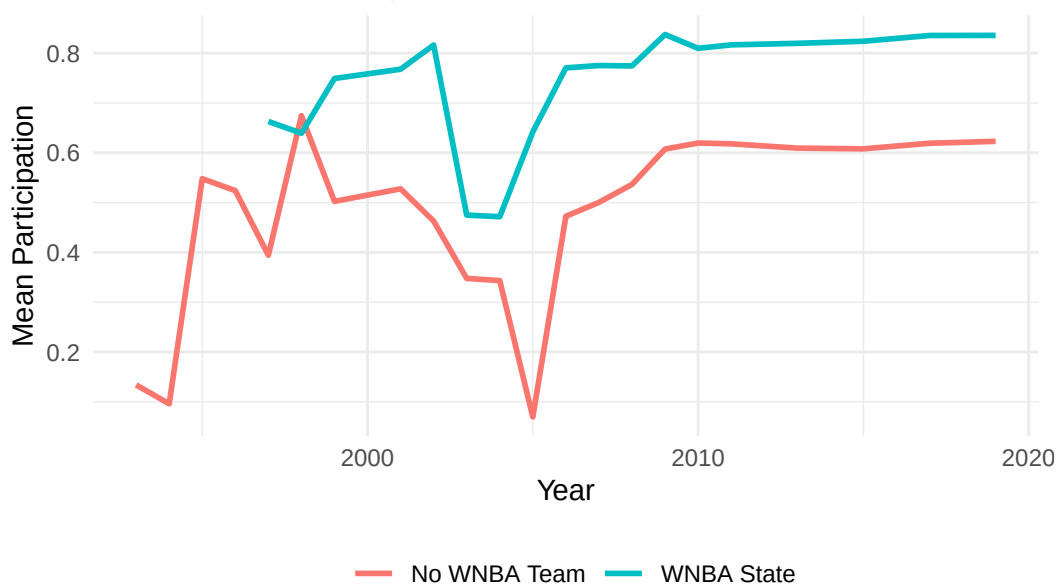
```

Girls Basketball Participation (Share of Total Participation): WN



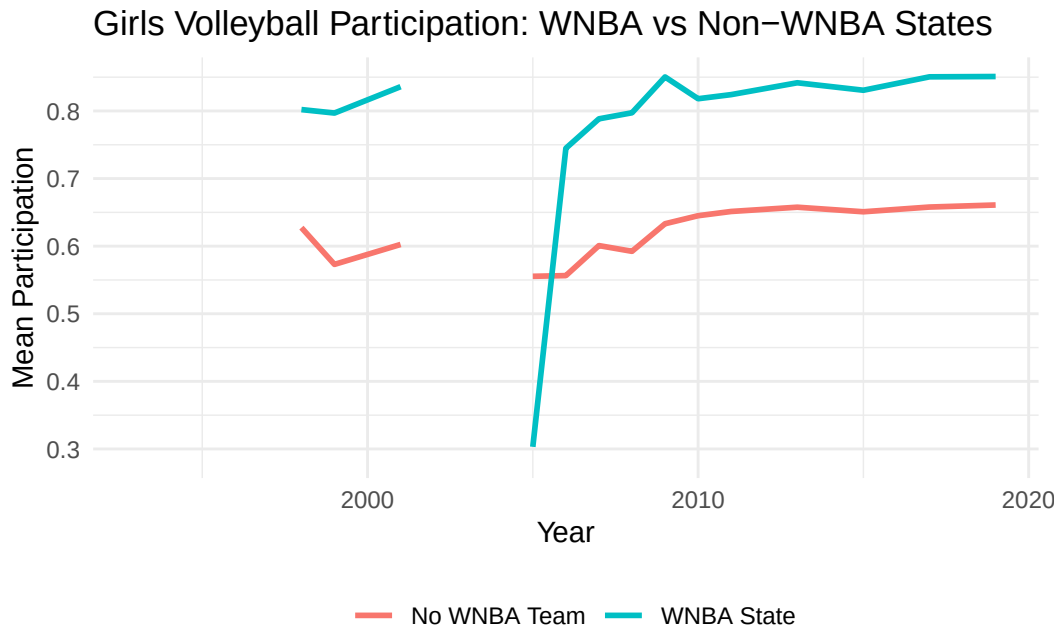
```
soc_df <- df_nfhs_wide %>%
  filter(gender == 1) %>%
  select(state, year, Soccer) %>%
  mutate(total = rowSums(across(where(is.numeric)), na.rm = TRUE)) %>%
  mutate(soccer_share = Soccer / total) %>%
  left_join(wnba_states %>% mutate(has_wnba = 1),
            by = c("state" = "state_num", "year")) %>%
  mutate(has_wnba = replace_na(has_wnba, 0),
         group = ifelse(has_wnba == 1, "WNBA State", "No WNBA Team")) %>%
  group_by(group, year) %>%
  summarise(participation_rate = mean(`soccer_share`, na.rm = TRUE), .groups = "drop") %>%
  filter(!is.na(year))
ggplot(soc_df, aes(x = year, y = participation_rate, color = group, group = group)) +
  geom_line(linewidth = 1) +
  labs(title = "Girls Soccer Participation: WNBA vs Non-WNBA States",
       x = "Year", y = "Mean Participation", color = "") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

Girls Soccer Participation: WNBA vs Non-WNBA States



```
vb_df <- df_nfhs_wide %>%
  filter(gender == 1) %>%
  select(state, year, Volleyball) %>%
  mutate(total = rowSums(across(where(is.numeric)), na.rm = TRUE)) %>%
  mutate(vb_share = Volleyball / total) %>%
  left_join(wnba_states %>% mutate(has_wnba = 1),
    by = c("state" = "state_num", "year")) %>%
  mutate(has_wnba = replace_na(has_wnba, 0),
    group = ifelse(has_wnba == 1, "WNBA State", "No WNBA Team")) %>%
  group_by(group, year) %>%
  summarise(participation_rate = mean(vb_share, na.rm = TRUE), .groups = "drop") %>%
  filter(!is.na(year))
ggplot(vb_df, aes(x = year, y = participation_rate, color = group, group = group)) +
  geom_line(linewidth = 1) +
  labs(title = "Girls Volleyball Participation: WNBA vs Non-WNBA States",
    x = "Year", y = "Mean Participation", color = "") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

Warning: Removed 3 rows containing missing values or values outside the scale range (`geom\_line()`).



It appears that having a WNBA team in the state leads to a higher participation rate in sports like basketball, soccer, and volleyball. There are many confounders here. While population size is one, that is at least partially controlled for by looking at the share of participation. Still, there are many other factors that could be at play, such as the general culture of sports in a given state, the presence of other sports teams, and the socioeconomic status of the population. A more rigorous analysis would be needed to establish causality. To do so, we implement a difference-in-differences regression, controlling for state and year fixed effects, and clustering standard errors at the state level. This would allow us to better isolate the effect of having a WNBA team on girls sports participation, while controlling for unobserved heterogeneity across states and over time.

DiD

$$Y_{st} = \alpha_0 + \alpha_1 \{ \text{Have WNBA} \times \text{Year} \} + \alpha_s + \alpha_t + \epsilon_{st}$$

```
library(did)
```

Warning: package 'did' was built under R version 4.5.2

Attaching package: 'did'

The following object is masked from 'package:psych':

sim

```
# Create treatment variable
df_did <- df_nfhs_wide %>%
  filter(gender == 1) %>%
  left_join(
    wnba_states %>% group_by(state_num) %>% summarise(first_treat = min(year)),
    by = c("state" = "state_num")
  ) %>%
  mutate(first_treat = replace_na(first_treat, 0),
         has_wnba = as.integer(first_treat > 0 & year >= first_treat))

library(lfe)
```

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

```
did_model <- feIbm(Basketball ~ has_wnba | state + year | 0 | state,
                  data = df_did)
summary(did_model)
```

Call:

```
feIbm(formula = Basketball ~ has_wnba | state + year | 0 | state,      data = df_did)
```

Residuals:

Min	1Q	Median	3Q	Max
-56410	-600	-96	501	56251

Coefficients:

	Estimate	Cluster s.e.	t value	Pr(> t)
has_wnba	1128	1154	0.978	0.333

Residual standard error: 4788 on 763 degrees of freedom
(379 observations deleted due to missingness)
Multiple R-squared(full model): 0.7337 Adjusted R-squared: 0.7089
Multiple R-squared(proj model): 0.002546 Adjusted R-squared: -0.09027
F-statistic(full model, *iid*):29.61 on 71 and 763 DF, p-value: < 2.2e-16
F-statistic(proj model): 0.9557 on 1 and 49 DF, p-value: 0.3331